

MB 510: Probability and Statistics for Data Science and AI

Course Code	MB 510
Course Name	Probability and Statistics for Data Science and AI
L-T-P-C	2-0-0-2 (L-T-P-C)
Prerequisites	None
Intended for	MBA
Distribution	Compulsory
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Uncertainty and probability, Probability and its types, Conditional, joint and marginal probability, Problem classification with emphasis on random problems, Probability and statistics concepts for DSAI. (4)
- **Random Variables and Probability Distributions:** Describing Randomness, Random Variables and Probability Distributions- Continuous and Discrete Distributions, Normal distribution, lognormal distribution and Power-Law Distributions, Bernoulli distributions. (4)
- **Statistics:** Collections of Random Values-Expected Value, variance and standard deviation, Independent and Identically Distributed Variables, law of large numbers, central limit theorem. (4)
- **Sampling:** Sampling and Replacement-Selection with Replacement, Selection Without Replacement, Bootstrapping. (4)
- **Bivariate Statistics:** Covariance and Correlation-Pearson and Spearman. (4)
- **Baye's Rule:** Frequentist vs. Bayesian Probability, Bays Rule and Confusion Matrices, Repeating Bayes' Rule, Multiple Hypothesis. (4)
- **Information Theory:** Entropy, Measuring uncertainty, information, and surprise, Maximal entropy distribution, Cross Entropy, KL Divergence (4)

Text Books

1. Nil

References

1. Thomas Nield, **Essential Math for Data Science: Take Control of Your Data with Fundamental Linear Algebra, Probability, and Statistics**, Shroff Publishers & Distributors Pvt. Ltd., Mumbai, 2022
2. Trevor Hastie, Robert Tibshirani, Jerome Friedman, **The Elements of Statistical Learning: Data Mining, Inference, And Prediction**, 2nd Edition, Springer, 2017.
3. Anirban Das Gupta, **Probability for Statistics and Machine Learning: Fundamentals and Advanced Topics**, Springer, 2011.
4. Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong, **Mathematics for Machine Learning**, Cambridge University Press (<https://mml-book.com>)
5. Scott E. Page, **The Model Thinker: What You Need to Know to Make Data Work for You**, Basic Books, 2018.
6. MS Excel 2019, **Data Analysis & Business Modeling**, Wayne Winston, Microsoft Press (PHI), 2020.